STRUCTURE & FUNCTIONS OF CYBER-PHYSICAL SYSTEMS

Risk Management Field Manual

Memorize the whole deck the easy way: one story, the exact exam points, and a memory hook on every topic.

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THD · NuW · AI for Smart Sensors

🤖 Meet Riku — your memory anchor

You and your friends are building **Riku, a self-driving delivery robot**. You're terrified it'll crash, hurt someone, or arrive late.

Every single slide just answers one question: **"How do clever engineers stop Riku from going wrong — as early and cheaply as possible?"** Keep Riku in your head and the chapter tells itself.

★ MASTER KEY → "Detectives Always Eat Cold Mexican Donuts" = Detect · Analyse · Evaluate · Control · Monitor · Document

00 · THE SHAPE OF THE WHOLE CHAPTER

See the skeleton before the bones

Risk Management is **step 3 of System Engineering**:

```
1 Basics → 2 Requirements → 3 RISK ★ → 4 Change → 5 Ideation → (6 Design) →
(7 Test) → (8 Project Setup)
```

The chapter solves risk in a fixed order. Learn this one silly sentence and you already own the structure of the exam — everything else just bolts onto one of these six words:

> 🧠 **"Detectives Always Eat Cold Mexican Donuts."**
> **D**etect → **A**nalyse → **E**valuate → **C**ontrol → **M**onitor → **D**ocument.

THE FULL LESSON · ALL SLIDES

Taught like a story — with the points you must write

01 What *is* a risk?

🧑 STORY

Before Riku even moves you ask, "what could go wrong?" The battery *might* die; it *might* rain. Hasn't happened yet → those are **risks**. The instant Riku actually crashes, it stops being a risk and becomes a **problem** — too late to prevent.

📝 WRITE IN EXAM — DEFINITION OF RISK

- "Risk is the effect of uncertainty" — definition from ISO 9001.
- RM's job: sort & structure risks in a structured way (→ PDCA).
- Risks = events/circumstances that can prevent reaching the project goal (probability < 100%).
- A risk that already occurred → a "problem" (treated differently).
- Separate general company risks from project-specific risks.

🧠 Risk = "not yet" · Problem = "already". Probability under 100% = still a risk.

02 PDCA — the engineer's heartbeat

🧑 STORY

How do you do anything well? Like baking cookies: **Plan** the recipe → **Do** the baking → **Check** the taste → **Act** to improve the next batch. Engineers spin this loop forever. From **ISO 9001**; also the **Deming** / **Shewhart** cycle.

📝 WRITE IN EXAM — PDCA

- Plan — plan process/product; set goals & resources; address risks & opportunities.
- Do — implement the plan.
- Check — monitor & measure the process / target condition.
- Act — take action to improve performance.
- Method used in “Plan”: FMEA.

🧠 P-D-C-A = “Please Don't Crash Again.” It's a circle, never a straight line.

03 Three clever truths (general principles)

🧒 STORY

Three surprising facts examiners love: (1) risk has a good twin — an **opportunity**; (2) people fear reporting bad news, so build an **anonymous “speak-up line”**; (3) money adds up but **risks don't** — you must combine them with maths like **Monte Carlo**.

📝 WRITE IN EXAM

- Opportunities can be recorded as positive events.
- Set up an anonymous “speak-up line” for undiscovered risks/violations.
- Risks do not add up → aggregate via models & simulations (Monte Carlo).

🧠 O-S-M → Opportunity · Speak-up · Monte-Carlo (risks don't add).

04 Don't put all eggs in one basket (diversification)

🧒 STORY

Bet all your pocket money on one lottery ticket = super risky. Spread it across different, *unrelated* things = safer for the same reward. Companies do the same with projects. To make Riku's 50%-reliable crash sensor safer, you must add something **different**, not a copy.

📝 **WRITE IN EXAM**

- Diversify — *"not all in one basket"*: **asset** (different products) & **time** diversification (idea by **Markowitz**, Nobel Prize).
- Optimum portfolios lie **on the efficiency curve**; below it = **inefficient**.
- To reduce risk: **diversified & non-correlating** solutions.
- ECU example — ❌ **wrong:** add a 2nd **identical** ECU (increases failure). ✅ **right:** add a **mechanical** crash detector.

🧠 Two of the **same** = double trouble. Two **different** = real safety.

05 The rulebook (standards)

👦 STORY

You can't just *say* you manage risk — big standards *force* you. **ISO 9001** demands "risk-based thinking"; **ISO 31000** is the dedicated risk standard; companies keep a money-jar (**provisions**) to pay if risks hit; and the **boss** is responsible.

📝 **WRITE IN EXAM**

- **ISO 9001** → **"risk-based thinking"** (use opportunities, prevent unwanted outcomes).
- **ISO 31000** → "Risk Management".
- Form **provisions (Rückstellungen)** for company & product risks → linked to **quality management**.
- Company-level RM = a **managerial task**.

🧠 **"9001 thinks, 31000 manages, the jar pays, the boss owns it."**

06 Risks come from everywhere (levels affected)

👦 STORY

Riku can fail in many directions. Turn 9 scary items into **3 buckets**: Product · Money/Time · People/Planet.

📝 **WRITE IN EXAM — 9 LEVELS**

- **Product:** Technical/product · Compliance
- **Money/Time:** Finances/costs · Time · Market · Currency
- **People/Planet:** Social · Project-management · Environmental
- Also: 6 levels of risk management by Gleißner & Mott; German law KonTraG forces management to do risk analysis.

🧠 3 buckets → Product · Money/Time · People/Planet. Remember the buckets, the items fall out.

07 Targets of Risk Management (master list)

🧒 STORY

This is the **D-A-E-C-M-D** skeleton, now with **Control** opened up. Controlling a risk = exactly 4 choices, like dealing with rain: **Avoid** (stay in), **Prevent** (umbrella), **Transfer** (send your brother), **Accept** (get a bit wet).

📝 **WRITE IN EXAM**

- **Targets:** Detect → Analyse → Evaluate → Control → Monitor → Document & communicate.
- **Control options:** Avoid · Prevent · Transfer · Accept.

🧠 Skeleton = "Detectives Always Eat Cold Mexican Donuts." Control = A-P-T-A (rain). Matches the classic 4 T's: Terminate, Treat, Transfer, Tolerate.

08 DETECT — finding risks

🧒 STORY

First, hunt for risks **regularly & systematically**, looking *inside* and *outside* the company. A slow stable company checks rarely; a fast-changing one checks often.

📝 **WRITE IN EXAM**

- Regular & systematic **internal and external** analysis of developments/events/changes; **frequency depends on volatility**.
- **6 methods to detect:** KPI changes · LeLe (lessons learned) · Headstand/Flip-Flop · Delphi · Risk workshop · PAAG/HAZOP.
- **Safety detection methods:** ALARP · FMEA · DRBFM.

🧠 **"Kids Love Hunting Down Risky Problems"** → KPI · LeLe · Headstand · Delphi · Risk-workshop · PAAG.

08a Head-stand / Flip-Flop method

🧒 STORY

Stand the question on its head! Instead of "how do we make Riku succeed?" ask **"how could we make Riku fail as FAST as possible?"** — kids find this funnier, so more ideas pop out. Then flip the silly answers back into real safety actions.

📝 **4 STEPS**

- (1) **Invert the thesis** → (2) **find solutions** for the flipped thesis → (3) **invert those solutions** → (4) **define actions**.

08b Delphi method

🧒 STORY

Ask many experts — but keep them in **separate rooms** so nobody copies or bullies. Everyone answers **anonymously**; you reveal the big disagreements, ask again, repeat until they roughly agree, then take the **average**. (Named after the ancient Oracle of Delphi.)

📝 **6 STEPS**

- (1) leader explains topic & targets → (2) worksheet to each expert, **no discussion** → (3) **anonymous** hand-in + evaluate → (4) share **major deviations** + comments, redistribute → (5) **repeat 2–4** until consolidated → (6) take the **average**.

09 EVALUATE — scoring the risk

STORY

Every risk gets **two scores**, like rating a scary movie: **Severity** (how bad) and **Occurrence** (how likely). They say "occurrence," NOT "probability," because it's a *human guess*, not a calculation. Combine the two → a **risk matrix** telling you what to fear first.

WRITE IN EXAM

- Two properties per risk: **Severity** + **Occurrence/likelihood** (subjective; *not* "probability").
- Combine → **risk matrix** to **prioritise**; keep an updated **risk list** (PM / risk manager).
- Judge **impact** via: **historical data** · **probability distribution** · **Monte Carlo simulation**.

Severity × Occurrence = Risk Matrix. "How bad × how likely." Low number = mild · high = nasty · 0 = gone · top = already happened.

10 CONTROL — the 5 ways to handle it

STORY

Rain again, plus a bonus tool: **Avoid** (ban it, occurrence→0; use barriers like Poka Yoke), **Poka Yoke** (make wrong physically impossible — like a SIM that fits one way), **Prevent** (umbrella), **Transfer** (give it to who copes better), **Accept** (shrug — only if small).

WRITE IN EXAM

- **Avoidance:** reduce occurrence to **0** — ban activities, remove features, **technical barriers (Poka Yoke)**.
- **Poka Yoke:** Japanese **"avoid silly mistakes"**; immediate error detection; **key-lock principle**.
- **Prevention:** reduce occurrence OR mitigate severity, using QM methods (e.g. **PDCA**).
- **Transfer:** shift the risk to who handles it best — e.g. crash **ECU → chassis**.
- **Acceptance:** consciously accept — **only if occurrence AND impact are low**.

🧠 **Avoid · Poka-Yoke · Prevent · Transfer · Accept** = ban · mistake-proof · umbrella · chassis · shrug.

11 FMEA — the star of the exam ⭐

🧒 STORY

FMEA is a giant **"what could break, and what happens then?"** checklist that a team of experts fills in **early**, before Riku is even built — so failures get caught in advance.

📝 WRITE IN EXAM — FMEA BASICS

- **FMEA = Failure Mode and Effects Analysis** (Ger. *Fehlermöglichkeiten- und Einflussanalyse*) — a **reliability-improvement method**.
- Rates defects by **Severity, Probability, Recognizability**.
- **Goals:** spot **systematic & random errors before they occur** · derive **detection & prevention** · improve **successor products**.
- **Types:** System FMEA · **D-FMEA** (Design, customer focus) · **P-FMEA** (Process). D & P must be **linked** (critical design feature = an **SC** checked in the process).
- **Standards:** Automotive **ISO 26262** · Industry IEC 61508 / ISO 13849 · Aerospace DO-254 / ARP 4761.
- **Process:** prepared **early**, in **expert workshops**, built **from cause → error**, then **verify the assumed probabilities**.
- **Living document:** maintain until **EOP** (better **EOS**); **update on every product change** (classic field-failure cause!); old FMEA feeds next project (**LeLe**).

🧠 Family: **A → +C(riticality)=FMECA → +D(iagnostics)=FMEDA.** "Add a letter to upgrade." · Standards anchor: **"cars = 26262."**

11a Rule of Ten 🔟

🧒 STORY

The longer a mistake hides, the more it costs — **×10 at every stage.** Catch Riku's bug while planning = 1€; let the customer find it = 1000€+. So fix things EARLY.

📝 **WRITE IN EXAM**

- Cost ×10 per stage: **Planning 1€ → Development 10€ → Assembly 100€ → Final acceptance 1000€ → Customer**. Focus on **planning & development**.

🧠 Every stage you're late, **add a zero**: 1 → 10 → 100 → 1000.

11b FMECA & FMEDA

📝 **WRITE IN EXAM**

- **FMECA** = FMEA + **Criticality** metric → enables **prioritisation** (usually what "FMEA" means in practice).
- **FMEDA** = FMEA + **Diagnostics**; **multiple-error analysis**; automotive electronics; counts how **self-detection/correction** lowers the failure rate; adds **quantifiable error data**; good for **complex systems**.

12 RPV — the FMEA score (S × O × R)

🧒 STORY

To rank each failure, FMEA gives **three marks out of 10** and multiplies them: how **bad** (Severity), how **often** (Occurrence), can our design **catch** it (Recognizability). Big number = fix first.

📝 **WRITE IN EXAM**

- **RPV = Severity × Probability(Occurrence) × Recognizability**, each rated **1–10**.
- Watch the flip: Recognizability **1 = best (≥99% detected)**, **10 = worst (<10% detected)**.

🧠 **"S × O × R" = "how SORe will it be?"** = how Bad × how Often × can we Recognize. (Classic FMEA **S×O×D = RPN**; Recognizability = Detection.)

12a Special Characteristic (SC)

STORY

If a feature is *super severe*, stamp it **SC** on the drawing — a red “MUST CHECK” sticker. Every SC is measured in production. But don't sticker *everything* (too costly).

WRITE IN EXAM

- High severity → **Special Characteristic (SC)**, marked on drawings, **each SC checked/measured**; can be a **legal/regulatory** requirement; don't over-use (cost!).
- **Automotive variants: BMW** = L/S/F (Legal/Safety/Function) · **Daimler** = DS/DZ · **VW** = D/S (Documented/Safety).

SC = red **“MUST-MEASURE”** sticker. BMW has **3 letters (LSF)**, VW has **2 (DS)**.

13 Loose ends + Summary

WRITE IN EXAM

- **Software** to track risks: **ClickUp · cplace · Jira** → key point: **consistent linking** with up/downstream processes.
- Worked **example:** bicycle **cable-pull** FMEA (*341_Example_FMEA_Bicycle*).
- **Summary:** RM **avoids greater damage** · identify risks **as early as possible** (Rule of Ten) · rate by **severity & occurrence** in a **risk list** · analyse products with **FMEA** & derive **SCs**.

Walter Scheel: “Nothing happens without risk, but without taking risks also nothing happens.” → manage risk, but still **take risks to move forward**.

YOUR PROGRESS

0 / 13 topics learned

ACTIVE RECALL · FLIP TO TEST YOURSELF

Flashcards

Read the question, say the answer *out loud*, then flip. Cover the deck and run it twice — that's how it sticks.

All flashcards (Q/A)

1. What is the definition of "risk" (ISO 9001)?
"Risk is the effect of uncertainty." An event that can prevent reaching the project goal (probability < 100%).

2. Risk vs problem?
Risk = NOT happened yet (<100%). Problem = ALREADY occurred — handled differently.

3. Two kinds of risk to keep separate?
General company risks vs project-specific risks.

4. PDCA + other names?
Plan–Do–Check–Act. Also the Deming / Shewhart cycle (ISO 9001).

5. PDCA — Plan?
Plan process/product, set goals & resources, address risks & opportunities.

6. PDCA — Do / Check / Act?
Do = implement; Check = monitor & measure; Act = improve performance.

7. Method used in the PDCA Plan phase?
FMEA.

8. Can a risk be positive?
Yes — opportunities recorded as positive events.

9. What is a speak-up line?
An anonymous channel to report undiscovered risks & violations.

10. Do risks add up like costs?
No — aggregate them with models/simulations (Monte Carlo).

11. Reduce risk through diversification?
Diversified & NON-correlating solutions; asset & time diversification (Markowitz).

12. 50% ECU — wrong vs right fix?
Wrong: 2nd identical ECU (raises failure). Right: a different, mechanical detector.

13. Efficiency curve meaning?
Optimum portfolios on the curve; none above; below = inefficient.

14. ISO 9001 risk concept?
"Risk-based thinking" — use opportunities, prevent unwanted outcomes.

15. ISO 31000?
The dedicated Risk Management standard.

16. Provisions (Rückstellungen)?
Money set aside for company/product risks; linked to quality management.

17. Company-level RM is what task?
A managerial task.

18. 9 levels risks affect?
Technical, Compliance, Finances, Time, Market, Currency, Social, Project-management, Environmental.

19. Who defined the 6 levels of RM?
Gleißner & Mott.

20. KonTraG?
German law obliging management to perform risk analysis.

21. The 6 targets/steps of RM?
Detect → Analyse → Evaluate → Control → Monitor → Document & communicate.

22. 4 options when CONTROLLING a risk?
Avoid · Prevent · Transfer · Accept.

23. What does Detect involve?
Regular & systematic internal AND external analysis; frequency depends on volatility.

24. 6 methods to detect risks?
KPI changes, LeLe, Headstand/Flip-Flop, Delphi, Risk workshop, PAAG/HAZOP.

25. 3 safety detection methods?
ALARP, FMEA, DRBFM.

26. Head-stand method — 4 steps?
Invert thesis → find solutions → invert solutions → define actions.

27. Delphi method key idea?
Experts answer anonymously, no discussion, iterate on deviations, take the average.

28. Evaluate: which two properties?
Severity (how bad) + Occurrence/likelihood (how likely; subjective, not 'probability').

29. Severity × Occurrence gives?
A risk matrix to prioritise; risks kept in a risk list (PM/risk manager).

30. 3 ways to assess impact?
Historical data, probability distribution, Monte Carlo simulation.

31. Severity & Occurrence scale ranges?
Severity 1(low)→4(very high). Occurrence 0(0%)→5(100%=occurred).

32. Risk Avoidance?
Reduce occurrence to 0 — ban activities, remove features, Poka Yoke barriers.

33. Poka Yoke?
Japanese 'avoid silly mistakes' — mistake-proofing; key-lock principle.

34. Risk Prevention?
Reduce occurrence OR severity using QM methods (PDCA).

35. Risk Transfer?
Shift the risk to who handles it best — e.g. crash ECU → chassis.

36. Risk Acceptance?
Consciously accept — only if occurrence AND impact are low.

37. FMEA — full form & what?
Failure Mode and Effects Analysis — a reliability-improvement method.

38. FMEA rates defects by?
Severity, Probability (Occurrence), Recognizability.

39. FMEA goals?
Spot systematic & random errors before they occur; derive detection & prevention; improve successor products.

40. FMECA?
FMEA + Criticality metric → enables prioritisation.

41. FMEDA?
FMEA + Diagnostics; multiple-error analysis; automotive electronics.

42. 3 types of FMEA?
System, D-FMEA (Design), P-FMEA (Process).

43. D-FMEA vs P-FMEA?
D = design/customer focus (avoid systemic errors); P = production-process weaknesses; must be linked.

44. FMEA standards by domain?
Automotive ISO 26262; Industry IEC 61508 / ISO 13849; Aerospace DO-254 / ARP 4761.

45. FMEA process?
Early, in expert workshops, from cause→error, then verify assumed probabilities.

46. Rule of Ten?
Cost ×10 per stage: Planning 1€→Dev 10€→Assembly 100€→Final 1000€→Customer.

47. FMEA maintained until?
EOP (better EOS); update on product changes; old FMEA feeds next project (LeLe).

48. How is RPV calculated?
RPV = Severity × Probability(Occurrence) × Recognizability, each 1–10.

49. Recognizability scale direction?

1 = surely recognized (≥99%); 10 = certainly not recognized (<10%).

50. Special Characteristic (SC)?

High-severity feature marked on drawings; each SC measured in production; can be legal.

51. SC automotive variants?

BMW L/S/F; Daimler DS/DZ; VW D/S.

52. System FMEA?

Whole system + subsystems; focuses on interfaces; finds random & systemic errors.

53. Software to track risks?

ClickUp, cplace, Jira — with consistent linking up/downstream.

54. Summary core message?

Avoid damage; identify early (Rule of Ten); rate by severity & occurrence; use FMEA + SC.

55. Walter Scheel quote?

"Nothing happens without risk, but without taking risks also nothing happens."

EXAM REHEARSAL · WRITE FIRST, THEN REVEAL

3 Practice Questions + model answers

Try to answer on paper before clicking **Reveal**. The model answers show the exact points an examiner rewards.

Q1 **Describe the risk-management process. Name and explain its main steps, and the options available when *controlling* a risk.**

TYPE: STRUCTURED / THEORY · ~10 MARKS

REVEAL MODEL ANSWER

✓ MODEL ANSWER

Risk management follows a fixed cycle: **Detect** → **Analyse** → **Evaluate** → **Control** → **Monitor** → **Document & communicate** (underpinned by the **PDCA** / Deming cycle from ISO 9001).

- **Detect:** regular & systematic internal and external analysis (KPI changes, lessons learned, head-stand method, Delphi, risk workshops, PAAG/HAZOP).
- **Analyse & Evaluate:** score each risk by **severity** and **occurrence/likelihood** and place it in a **risk matrix** / risk list to prioritise.
- **Control** — four options: **Avoid** (reduce occurrence to 0, e.g. Poka Yoke), **Prevent** (reduce occurrence or severity, e.g. via PDCA), **Transfer** (shift to who handles it best, e.g. ECU→chassis), **Accept** (consciously, only if occurrence and impact are low).
- **Monitor:** check whether risks materialise; **Document & communicate** and define the RM organisation.

Distinguish a **risk** (not yet occurred, <100%) from a **problem** (already occurred). Note risks **do not add up** — aggregate via models (Monte Carlo).

Q2

Explain FMEA. Distinguish FMEA, FMECA and FMEDA, state how the Risk Priority Value (RPV) is calculated, and explain the "Rule of Ten".

TYPE: THEORY / DEFINITIONS · ~12 MARKS

REVEAL MODEL ANSWER

✓ MODEL ANSWER

FMEA (**Failure Mode and Effects Analysis**) is a reliability-improvement method that, early and in cross-domain expert workshops, lists how a product can fail and what the effects are. It works **from cause to error** and aims to recognise systematic & random errors **before they occur**, derive detection & prevention measures, and improve successor products. It is a **living document** maintained until EOP/EOS.

- **FMEA → FMECA:** adds a **Criticality** metric → enables prioritisation.
- **FMEA → FMEDA:** adds **Diagnostics** (multiple-error analysis, automotive electronics) — credits self-detection that lowers the failure rate.

RPV = **Severity** × **Probability**(**Occurrence**) × **Recognizability**, each rated 1–10 (for Recognizability, 1 = surely detected, 10 = not detected). A high RPV means act first.

Rule of Ten: the later an error is found, the higher the cost — roughly ×**10 per stage** (Planning 1€ → Development 10€ → Assembly 100€ → Final acceptance 1000€ → Customer). Therefore effort must focus on **planning & development**.

Q3 **Riku, an autonomous delivery robot, has an obstacle-detection sensor that is only 50% reliable. (a) Use diversification to improve it — and say what would be WRONG. (b) Classify two risks with severity & occurrence. (c) Name two ways to detect risks.**

TYPE: APPLICATION · ~8 MARKS

REVEAL MODEL ANSWER

✓ **MODEL ANSWER**

(**a**) Reliability is improved by **diversified, non-correlating** solutions. **Wrong:** add a second **identical** sensor — same solution to the same problem; it even raises the failure rate. **Right:** add a **different** detection principle (e.g. a mechanical/ultrasonic detector) so the two don't fail together. (Markowitz — don't put all eggs in one basket.)

(**b**) Examples (Severity 1–4, Occurrence 0–5 → risk matrix):

- Sensor briefly blinded by sunlight → frequent but mild: **Occurrence high**, **Severity low** → **medium**.
- Robot fails to brake and hits a pedestrian → rare but catastrophic: **Occurrence low**, **Severity very high** → **high**.

(**c**) Any two of: **Head-stand/Flip-Flop method**, **Delphi method**, KPI-change monitoring, lessons learned (LeLe), risk workshop, PAAG/HAZOP, or safety methods ALARP / FMEA / DRBFM. *(Head-stand: ask "how could we make Riku crash as fast as possible?", then invert the answers into safety actions.)*

LAST 5 MINUTES BEFORE THE EXAM

One-screen cheat-card

Skeleton	**Detect·Analyse·Evaluate·Control·Monitor·Document**	“Detectives Always Eat Cold Mexican Donuts”
Control 4	**Avoid·Prevent·Transfer·Accept**	Rain: A-P-T-A (= 4 T's)
Risk vs problem	**not-yet vs already**	<100% = risk
PDCA	**Plan·Do·Check·Act**	“Please Don't Crash Again” (circle)
Detect methods	**KPI·LeLe·Headstand·Delphi·Risk-workshop·PAAG**	“Kids Love Hunting Down Risky Problems”
Evaluate	**Severity × Occurrence → matrix**	bad × likely
FMEA family	**A → +C(rit) → +D(iag)**	add a letter to upgrade
RPV	**Severity × Occurrence × Recognizability**	“how SORe” (= S×O×D)
Rule of Ten	**×10/stage, 1→1000€**	each stage late = +1 zero
Diversify	**different & non-correlating**	2 same = double trouble
SC	**high severity → must-measure**	BMW=LSF, VW=DS
Standards	**cars = ISO 26262**	9001 thinks · 31000 manages
Head-stand	**flip → solve → flip back → act**	stand it on its head
Delphi	**silent · anonymous · repeat · average**	Oracle of Delphi
Levels affected	**Product / Money-Time / People-Planet**	3 buckets

Companion files: 3.md · 3_memorize.md · 3_flashcards.txt · 3_study.pdf